The First Pass: A Complete Worked Example Through the Generative-Collapse Spine

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This paper is the shortest complete route through the Generative Collapse Dynamics (GCD) validation spine. We trace a single measurement—three voltage readings at 99% fidelity—from Axiom-0 ("Collapse is generative; only what returns is real") through every stop of the Universal Measurement Contract Protocol (UMCP): frozen contract, bounded embedding, kernel computation, integrity ledger, regime classification, and three-valued verdict. Every number is exact, every identity is verified to machine precision, and every step is reproducible from the open-source reference implementation. A heterogeneous counterexample exposes geometric slaughter—the mechanism by which a single degraded channel destroys multiplicative coherence while the arithmetic mean appears healthy. The paper requires no prior knowledge of the framework: a reader who has followed one complete pass can navigate any domain closure, any theorem, and any extension of the system.

I. INTRODUCTION

The Generative Collapse Dynamics (GCD) framework and its operational arm, the Universal Measurement Contract Protocol (UMCP) [1, 2], have been validated across 23 domain closures [3]—from particle physics [2] to consciousness coherence—with 20,221 automated tests, 47 lemmas, and 44 structural identities verified to machine precision. Yet no single document has traced one complete object through every layer of the system: from the axiom through the frozen contract, embedding, kernel computation, ledger reconciliation, regime classification, casepack packaging, and operational execution—all with exact numbers.

This paper provides that trace. The object is deliberately minimal: three identical voltage readings. This choice is architectural, not pedagogical laziness. The validation spine is *invariant* across domains: the same frozen contract, kernel, ledger, regime gates, and verdict logic apply whether the trace vector carries three voltages or thirty-one Standard Model particles. Only the Tier-2 closure changes—the mapping from raw observables to the unit interval. A reader who has verified the spine for the simplest case can navigate any domain closure by understanding only the channel selection, never the protocol itself. A *closure* is precisely this mapping layer: it selects which real-world quantities become the trace vector's channels and weights, but does not alter the kernel or the protocol that evaluates them.

A. Scope and structure

We present twelve stops, organized into the five stages of the fixed discourse spine:

1. **Contract** (Stops 1–2): The axiom is stated; the contract is frozen.
2. **Canon** (Stops 3–5): Channels are chosen, the trace is bounded, and the kernel is computed.
3. **Closures** (Stop 6): Thresholds are published; the ledger is debited and credited.
4. **Integrity Ledger** (Stop 7): The seam residual is checked; the regime is classified.
5. **Stance** (Stops 8–12): The verdict is derived; the result is packaged, executed, rendered, and read back into ordinary language.

Figure 1 maps the twelve stops onto the five spine stages visually.

Stop	What happens	Spine stage	Tier
1	Axiom declared	Contract	Tier-1
2	Parameters frozen		Tier-0
3	Channels chosen	Canon	Tier-2
4	Trace bounded		Tier-0
5	Kernel computed		$1 \leftarrow 0$
6	Ledger debited / credited	Closures	Tier-0
7	Regime classified	Integrity Ledger	Tier-0
8	Casepack assembled	Stance	Tier-0
9	CLI executed		Tier-0
10	API called		Tier-0
11	Latin rendered		Tier-2
12	Ordinary reading		Tier-2

FIG. 1. Twelve stops mapped onto the five spine stages. Tier-1: kernel and its identities; Tier-0: protocol implementation; Tier-2: domain-specific channel selection. "$1 \leftarrow 0$": Tier-0 code evaluates a Tier-1 function.

B. Source files

The worked example traces through the following repository paths (relative to the project root):

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Role	Path
Casepack	casepacks/hello_world/
Contract	contracts/UMA.INTSTACK.v1.yaml
Kernel code	src/umcp/kernel_optimized.py
Seam budget	src/umcp/seam_optimized.py
Frozen params	src/umcp/frozen_contract.py
Validator	src/umcp/validator.py
CLI entry	src/umcp/cli.py
This guide	THE_FIRST_PASS.md

Appendix A provides a heterogeneous counterexample. Appendix B establishes the precise relationship between the GCD kernel and classical results.

II. THE AXIOM

Axiom (Axiom-0)**.** Collapse is generative; only what returns is real.

Collapsus generativus est; solum quod redit, reale est.

This is the single foundational principle. It is a constraint on admissible claims: no measurement, computation, or verdict is admitted unless it can demonstrate *return*—meaning the system can re-enter its admissible neighborhood D_θ after drift, perturbation, or delay, under the same evaluation rules that were declared before the data was seen.

Formally, return at time t requires the existence of a prior state $u \in D_\theta(t)$ such that $\|\Psi(t) - \Psi(u)\| \leq \eta$, where η is the frozen return tolerance. When return is indefinitely deferred, $\tau_R = \infty_{\text{rec}}$, and no epistemic credit is granted.

Everything that follows derives from this axiom.

III. THE FROZEN CONTRACT

Before touching data, we declare the rules. The contract UMA.INTSTACK.v1 [4] freezes all parameters prior to computation.

Definition 1 (Frozen parameters)**.** *The frozen parameters are:*

$$\begin{aligned} \varepsilon &= 10^{-8} && \textit{(guard band)}, \\ p &= 3 && \textit{(drift cost exponent)}, \\ \alpha &= 1.0 && \textit{(curvature cost coefficient)}, \\ \text{tol}_{\text{seam}} &= 0.005 && \textit{(seam residual tolerance)}. \end{aligned}$$

These values are not arbitrary. They are *seam-derived*: the unique constants where mathematical identities hold consistently across all 23 tested domains [2, 3]. Operationally:

- $\varepsilon = 10^{-8}$: no channel can fully die—this guard band keeps $\ln(c_i)$ finite without affecting any measurement to machine precision.
- $p = 3$: the unique integer for which ω_{trap} is a Cardano root of $x^3 + x - 1 = 0$, governing how steeply the drift cost $\Gamma(\omega)$ penalizes departure from fidelity.
- $\alpha = 1.0$: unit coupling—each unit of curvature contributes one unit of cost to the ledger, with no amplification or suppression.
- $\text{tol}_{\text{seam}} = 0.005$: the width at which $\text{IC} \leq F$ holds at 100% across every tested domain; a residual below this threshold means the seam closes.

The contract also declares three algebraic identities that every computation must satisfy:

$$F + \omega = 1 \qquad \text{(duality)}, \tag{1}$$

$$\text{IC} \leq F \qquad \text{(integrity bound)}, \tag{2}$$

$$\text{IC} = e^{\kappa} \qquad \text{(log-integrity)}. \tag{3}$$

And three-valued verdicts: CONFORMANT | NONCONFORMANT | NON-EVALUABLE. Never boolean.

IV. CHANNEL CHOICE AND EMBEDDING

At $t = 0$, a sensor reports three raw values:

$$x_1 = 9.9, \quad x_2 = 9.9, \quad x_3 = 9.9, \tag{4}$$

with declared instrument range $[0, 10]$.

Channel choice is a Tier-2 act: the domain closure maps physical quantities to the unit interval. With linear normalization:

$$c_k = \frac{x_k - x_{\min}}{x_{\max} - x_{\min}} = \frac{x_k}{10}, \tag{5}$$

yielding $c_1 = c_2 = c_3 = 0.99$. Equal weights $w_k = 1/3$ are assigned (no domain-specific reason to privilege any channel).

Definition 2 (Trace and weight vectors)**.** *The trace vector is* $\mathbf{c} = (0.99, 0.99, 0.99) \in [0, 1]^3$. *The weight vector is* $\mathbf{w} = (1/3, 1/3, 1/3) \in \Delta^2$. *Both satisfy the contract constraints:* $c_k \in [\varepsilon, 1 - \varepsilon]$ *and* $\sum_k w_k = 1$.

A. The guard band

The contract specifies face policy pre_clip: before any computation, every channel is clamped to $[\varepsilon, 1 - \varepsilon]$. For $c_k = 0.99$, the clamp is inactive. The guard band exists because the kernel requires $\ln(c_k)$; if $c_k = 0$, the kernel diverges. The ε-clamp is a return guarantee: no channel can fully die, because a dead channel has no path back through collapse.

If a raw value exceeds the declared bounds (e.g., $x_1 = 12.5$ when the range is $[0, 10]$), the embedding produces $c_1 = 1.25$, which clips to $c_1 = 1 - \varepsilon \approx 1.0$, and an out-of-range (OOR) flag is set. The data is preserved; the flag is the audit trail.

For our trace: no OOR flags. The bounded trace is

$$\Psi(0) = (0.99, 0.99, 0.99). \tag{6}$$

V. THE KERNEL

The kernel $K : [0,1]^n \times \Delta^{n-1} \to \mathbb{R}^6$ maps the trace and weight vectors to six invariants. We compute each explicitly.

Definition 3 (GCD Kernel)**.** *Given* $\mathbf{c} = (c_1, \ldots, c_n) \in [\varepsilon, 1-\varepsilon]^n$ *and* $\mathbf{w} = (w_1, \ldots, w_n)$ *with* $\sum_i w_i = 1$*:*

$$F = \sum_i w_i\, c_i, \tag{7}$$

$$\omega = 1 - F, \tag{8}$$

$$\kappa = \sum_i w_i \ln c_i, \tag{9}$$

$$\mathrm{IC} = e^{\kappa}, \tag{10}$$

$$S = -\sum_i w_i \big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\big], \tag{11}$$

$$C = \frac{\mathrm{stddev}(c_i)}{0.5}. \tag{12}$$

A. Fidelity and drift

Fidelitas (F) measures what survives collapse:

$$F = \tfrac{1}{3}(0.99 + 0.99 + 0.99) = 0.99. \tag{13}$$

Derivatio (ω) measures what is lost:

$$\omega = 1 - 0.99 = 0.01. \tag{14}$$

Proposition 4 (Duality verification)**.** $F + \omega = 0.99 + 0.01 = 1.0$ *exactly. The residual is* $|F + \omega - 1| = 0.0$.

This is not approximate. The duality identity (1) holds to 0.0×10^0 residual across all tested traces [3].

B. Log-integrity and integrity

Log-Integritas (κ):

$$\kappa = \tfrac{1}{3}\big[\ln(0.99) + \ln(0.99) + \ln(0.99)\big] = -0.01005034. \tag{15}$$

$\kappa \le 0$ always, since $\ln(c_i) \le 0$ for $c_i \in (0,1]$.

Integritas Composita (IC):

$$\mathrm{IC} = e^{-0.01005034} = 0.99. \tag{16}$$

The log-integrity relation (3) holds by construction.

Proposition 5 (Integrity bound verification)**.** $\mathrm{IC} = 0.99 \le 0.99 = F$. *For homogeneous traces* ($c_1 = c_2 = \cdots = c_n$)*, equality holds:* $\mathrm{IC} = F$.

The integrity bound (2) is the solvability condition for trace recovery: given (F, IC) with $n = 2$, the individual channels satisfy $c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}$, which requires $\mathrm{IC} \le F$ for real solutions. This is derived independently from Axiom-0; the classical AM-GM inequality is the degenerate limit when channel semantics, weights, and the guard band are removed.

C. Entropy

The Bernoulli field entropy (not Shannon entropy, which is the degenerate limit when the collapse field is set to $\{0,1\}$):

$$\begin{aligned} S &= -\tfrac{1}{3}\big[0.99\ln(0.99) + 0.01\ln(0.01)\big] \times 3 \\ &= 0.05600. \end{aligned} \tag{17}$$

This is low: the channels are near-maximal, leaving little uncertainty in the collapse field.

D. Curvature

Curvatura (C) measures channel spread:

$$C = \frac{\mathrm{stddev}(0.99, 0.99, 0.99)}{0.5} = \frac{0.0}{0.5} = 0.0. \tag{18}$$

With identical channels, there is no heterogeneity.

E. The heterogeneity gap

The gap $\Delta = F - \mathrm{IC}$ measures how much the arithmetic mean conceals about channel dispersion:

$$\Delta = 0.99 - 0.99 = 0.0. \tag{19}$$

For homogeneous traces, $\Delta = 0$. This will become nontrivial in Appendix A.

F. Kernel summary

TABLE I. Kernel outputs for $\mathbf{c} = (0.99, 0.99, 0.99)$ with equal weights.

Symbol	Latin	Value	Meaning
F	Fidelitas	0.99	What survives
ω	Derivatio	0.01	What is lost
κ	Log-Integritas	-0.01005	Log coherence
IC	Integritas	0.99	Mult. coherence
S	Entropia	0.05600	Field entropy
C	Curvatura	0.0	Channel spread
Δ	—	0.0	Heterogeneity gap

VI. THE INTEGRITY LEDGER

The integrity ledger tracks debits and credits across the seam [5]:

Definition 6 (Seam budget)**.** *The budget identity is*

$$\Delta\kappa = R \cdot \tau_R - (D_\omega + D_C), \tag{20}$$

where the drift cost is

$$D_\omega = \Gamma(\omega) = \frac{\omega^p}{1-\omega+\varepsilon} = \frac{(0.01)^3}{0.99+10^{-8}} = 1.01\times 10^{-6}, \tag{21}$$

and the curvature cost is

$$D_C = \alpha \cdot C = 1.0 \times 0.0 = 0.0. \tag{22}$$

At $t = 0$, there is no prior history. The system has never departed and never returned, so $\tau_R = \infty_{\text{rec}}$—infinite return delay. Under the contract's censoring rule (*Si* $\tau_R = \infty_{\text{rec}}$, *nulla fides datur*), no credit is earned:

$$R \cdot \tau_R = 0. \tag{23}$$

The ledger:

Entry	Type	Amount
Drift cost D_ω	Debit	1.01×10^{-6}
Curvature cost D_C	Debit	0.0
Return credit	Credit	0.0
Residual $\lvert s\rvert$		1.01×10^{-6}

The seam closes: $|s| = 1.01\times 10^{-6} \le 0.005 = \text{tol}_{\text{seam}}$.

Why $\tau_R = \infty_{\text{rec}}$ and the paper is still conformant

(1) **No return credit.** Because $\tau_R = \infty_{\text{rec}}$, the return credit $R\cdot\tau_R$ is exactly zero. The system has never departed and never returned, so the ledger earns nothing on the credit side.
(2) **The seam still closes.** Conformance requires only that the residual $|s| \le \text{tol}_{\text{seam}}$. For a near-perfect trace the debit side is negligible ($D_\omega \approx 10^{-6}$, $D_C = 0$), so the seam closes even with zero credit.
(3) ∞_{rec} **is a typed semantic value.** In data files it appears as the string `INF_REC`; in Python it maps to `float("inf")`. It is not an error, not a missing datum, and not a placeholder—it is the legitimate verdict "no return has occurred."
(4) **First-pass admissibility.** A first measurement has no history against which to measure return. The protocol handles this by construction: the ledger reconciles without credit, and the regime gates evaluate independently of τ_R. History begins here; return is measured from the next pass onward.

VII. REGIME CLASSIFICATION

The regime is classified by four conjunctive gates applied to the kernel outputs. These gates are frozen in the contract [4]:

Definition 7 (Regime gates)**.**

STABLE : $\omega < 0.038 \;\wedge\; F > 0.90 \;\wedge\; S < 0.15 \;\wedge\; C < 0.14$,
WATCH : $0.038 \le \omega < 0.30$ *(or Stable gates not all met)*,
COLLAPSE : $\omega \ge 0.30$,
CRITICAL : IC < 0.30 *(overlay)*.

For our trace:

Gate	Threshold	Value	Pass?
$\omega < 0.038$	0.038	0.01	✓
$F > 0.90$	0.90	0.99	✓
$S < 0.15$	0.15	0.056	✓
$C < 0.14$	0.14	0.0	✓

All four gates pass. Regime: STABLE. Critical overlay: inactive (IC $= 0.99 > 0.30$). Stance: CONFORMANT.

Remark 1. *In Fisher coordinate space ($\theta = \arcsin\sqrt{\omega}$), the* STABLE *regime occupies only 12.5% of the total manifold. The remaining 87.5% is* WATCH *(24.4%) and* COLLAPSE *(63.1%). Stability is structurally rare.*

The regime, the stance, and the validator result are distinct objects. Table II separates them.

TABLE II. Four related but distinct concepts. Each is computed from different inputs and answers a different question.

Object	Computed from	Meaning	This run
Regime	F, ω, S, C gates	Structural phase of the trace	Stable
Stance	Ledger + regime + identities	Three-valued verdict	Conf.
Validator	Schema, sums, IDs	Protocol compliance	Conf.
Return	τ_R from history	Re-entry into D_θ?	$\tau_R{=}\infty_{\text{rec}}$

VIII. OPERATIONAL EXECUTION

The preceding computation is packaged in a *casepack*—the atomic unit of auditable work [4]. A casepack is a directory containing every artifact needed to reproduce the computation and verify the verdict. The `hello_world` casepack contains:

```
casepacks/hello_world/
  manifest.json
  raw_measurements.csv
  closures/
    closure_registry.yaml
  contracts/
```

```
    contract.yaml
    embedding.yaml
    return.yaml
    weights.yaml
  expected/
    invariants.json
    psi.csv
    ss1m_receipt.json
```

- `manifest.json`: provenance metadata—timestamps, tool versions, SHA-256 checksums binding every artifact to a unique digest.
- `raw_measurements.csv`: the raw measurements (here, three voltages at 9.9).
- `contracts/`: the frozen contract files—contract terms, embedding config, return policy, and weight declarations.
- `closures/closure_registry.yaml`: lists which Tier-2 closures are active for this run.
- `expected/`: the expected outputs—embedded trace (`psi.csv`), kernel invariants (`invariants.json`), and the Single-Source Single-Measurement receipt (`ss1m_receipt.json`).

A. Command-line validation

```
$ umcp validate casepacks/hello_world \
    --strict
UMCP validation: CONFORMANT
  (repo + casepacks/hello_world),
  errors=0 warnings=0;
  validator=umcp-validator v2.3.1;
  policy strict=true;
  sha256=6fc79df4...
```

The validator performs, in sequence: JSON Schema validation (Draft 2020-12), semantic rule checks (15+ rules), kernel identity checks ($F = 1 - \omega$, $\mathrm{IC} \approx e^{\kappa}$, $\mathrm{IC} \leq F$), regime consistency verification, and SHA-256 integrity checking. Exit code: 0 = CONFORMANT, 1 = NONCONFORMANT.

B. Programmatic API

The same computation is available via the Python API:

```
from umcp import validate
result = validate("casepacks/hello_world")
assert result.status == "CONFORMANT"
```

Or via the `MeasurementEngine` for raw data (the embedding config declares the instrument range $[0, 10]$ from Eq. (4)):

```
from umcp.measurement_engine import (
    MeasurementEngine, EmbeddingConfig,
    EmbeddingSpec, EmbeddingStrategy)
import numpy as np
spec = EmbeddingSpec(
    strategy=
      EmbeddingStrategy.LINEAR_SCALE,
    input_range=(0, 10))
emb = EmbeddingConfig(specs=[spec] * 3)
engine = MeasurementEngine()
raw = np.array([[9.9, 9.9, 9.9]])
result = engine.from_array(
    raw, weights=[1/3, 1/3, 1/3],
    embedding=emb)
row = result.invariants[0]
# row.F == 0.99, row.regime == "STABLE"
```

C. REST endpoint

The optional FastAPI extension exposes the same spine over HTTP:

```
POST /validate HTTP/1.1
Content-Type: application/json

{"path": "casepacks/hello_world",
 "strict": true}

HTTP/1.1 200 OK

{"status": "CONFORMANT",
 "errors": 0,
 "warnings": 0,
 "path": "casepacks/hello_world",
 "strict": true,
 "created_utc": "2026-...",
 "hash": "6fc79df4..."}
```

All three interfaces—CLI, Python API, REST—traverse the same frozen spine; no interface can produce a verdict that bypasses the kernel or the contract.

D. The cognitive equalizer (Aequator Cognitivus)

Every execution path—CLI, Python API, REST endpoint, C99 core, C++ accelerator—passes through the same spine with the same frozen parameters and produces the same verdict. This is the *Aequator Cognitivus*: the system produces identical output regardless of which cognitive agent operates it. Thresholds are frozen, vocabulary is operationally defined, and verdicts are derived from gates. The structure measures; the agent does not.

IX. LATIN RENDERING AND CROSS-DOMAIN TRANSLATION

The following section is not ornamentation. Each Latin term carries morphological constraints that English prose does not—the operational definition is encoded in the word's structure. We render the worked example in full canonical form to verify that the five-word vocabulary and the Rosetta translation are operational, not decorative.

The worked example, rendered canonically:

> *Ad tempus $t = 0$, tria vestigia mensurantur:* $\mathbf{c} = (0.99, 0.99, 0.99)$. *Fidelitas est* $F = 0.99$*; derivatio est* $\omega = 0.01$. *Complementum perfectum tenet:* $F + \omega = 1$. *Log-integritas est* $\kappa = -0.01005$*; integritas composita est* $\mathrm{IC} = 0.99$. *Limbus integritatis tenet:* $\mathrm{IC} \leq F$. *Entropia campi est* $S = 0.056$*; curvatura est* $C = 0.0$. *Moratio reditus est* $\tau_R = \infty_{\mathrm{rec}}$*—nulla fides datur. Regio est Stabilis. Sutura clauditur. Iudicium:* CONFORMANT.

A. The five-word vocabulary

The canonical prose interface uses five words, each with a ledger role [1]:

Latin	English	Value
Derivatio	Drift	$\omega = 0.01$
Fidelitas	Fidelity	$F = 0.99$
Curvatura	Roughness	$C = 0.0$
Moratio Reditus	Return	$\tau_R = \infty_{\mathrm{rec}}$
Integritas	Integrity	$\mathrm{IC} = 0.99$

B. Rosetta translation

The Rosetta adapter [1] maps the five words across lenses so different fields can read the same computation in their own dialect:

Lens	Drift = 0.01	Fidelity = 0.99
Physics	1% energy loss	99% preserved
Finance	1% drawdown	99% retained
Epistemology	Minimal shift	Warrant preserved

The meanings of the columns remain stable while the dialect changes. *Significatio stabilis manet dum dialectus mutatur.*

X. DEGENERATE LIMITS

The GCD kernel structures relate to classical results as follows: the classical result is what remains when degrees of freedom are removed. The arrow of derivation runs from the axiom to the classical limit, never in reverse.

Proposition 8 (Classical degenerate limits). *(i)* ***Duality → unitarity.*** *$F + \omega = 1$ reduces to $p + q = 1$ when the cost function $\Gamma(\omega)$ is removed.*

(ii) ***Integrity bound → AM-GM inequality.*** *$\mathrm{IC} \leq F$ reduces to $\mathrm{GM}(c) \leq \mathrm{AM}(c)$ when channel semantics, weights, and the guard band ε are removed.*

(iii) ***Bernoulli field entropy → Shannon entropy.*** *The entropy S of Eq. (11) reduces to Shannon entropy when the collapse field is removed (i.e., $c_i \in \{0, 1\}$ only).*

(iv) ***Log-integrity → exponential map.*** *$\mathrm{IC} = e^{\kappa}$ reduces to the standard exponential map when the weighted sum structure is removed.*

The GCD framework does not "use" or "apply" these classical results. It derives the kernel identities independently from Axiom-0; the classical results emerge as structurally impoverished special cases. These are structural containment claims internal to the formalism—they describe how the GCD kernel's degrees of freedom reduce to classical formulas when structure is removed, not broad historical priority claims about the classical results themselves.

XI. DISCUSSION

The twelve stops above constitute the shortest complete route through the GCD validation spine. Because the route is complete—every stop visited, every number exact—several structural properties become visible that a partial trace would not expose:

1. **Contract precedence.** All parameters are frozen before any data is seen. No threshold, tolerance, or gate is tuned after evidence.

2. **Exact identities.** $F + \omega = 1$ holds to 0.0×10^0 residual; $\mathrm{IC} = e^{\kappa}$ holds to machine precision; $\mathrm{IC} \leq F$ with equality for homogeneous traces. These are not checked *in addition to* the spine; they are the spine's own structural constraints.

3. **Three-valued logic.** The verdict is CONFORMANT, NONCONFORMANT, or NON-EVALUABLE—never boolean. The third state is revealed only when the full route is traversed, because a shortcut that omits the ledger or the regime would collapse the verdict space to two values.

4. **Typed boundaries.** $\tau_R = \infty_{\mathrm{rec}}$ is a first-class semantic value (not an error or a missing datum) that directly determines ledger credit. The first pass makes this visible: a measurement with no history earns zero credit and still conforms.

5. **Interface invariance.** CLI, Python API, REST API, C99 core, and C++ accelerator all produce identical results because each traverses the same frozen spine.

6. **Structural stability is rare.** STABLE occupies 12.5% of Fisher space; the homogeneous high-fidelity trace sits in the small corner where all four gates simultaneously pass.

Each point is a consequence of completeness: it could not be demonstrated without visiting every stop.

A. Reproduce this pass

Every number in this paper can be regenerated from the open-source repository [3]. The following commands suffice:

Run This Now

Clone & install

```
git clone https://github.com/
  calebpruett927/
  GENERATIVE-COLLAPSE-DYNAMICS.git
cd GENERATIVE-COLLAPSE-DYNAMICS
pip install -e ".[all]"
```

Validate the hello_world casepack

```
umcp validate casepacks/hello_world
    --strict
```

Run the orientation (10 sections)

```
python scripts/orientation.py
```

Run the full test suite

```
pytest
```

XII. CONCLUSIONS

The GCD validation spine has exactly one shortest complete route: freeze the contract, embed the data, compute the kernel, reconcile the ledger, classify the regime, derive the verdict. This paper has walked that route with exact numbers for the simplest possible input, proving by construction that every stop is necessary and sufficient. A reader who has followed this single pass can enter the framework at any point—any of 23 domain closures, any of the 44 structural identities, any extension—because the protocol that governs them is precisely what was traversed here.

The first pass is complete. Now proceed through any door.

Finis primi transitus. Nunc per quamlibet portam procede.

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This work was conducted independently without institutional affiliation or external funding. The kernel's mathematical substrate has a long history: Jakob Bernoulli's *Ars Conjectandi* (1713) established the field structure from which the entropy formula descends; R. A. Fisher's geometric treatment of statistical families (1925) provides the coordinate space in which regime partition is measured; Claude Shannon's channel capacity theorem (1948) is recovered here as a degenerate limit when the collapse field is removed; and the metrological tradition codified in the GUM [6] established the principle that measurement uncertainty must be declared before evidence. The frozen-contract discipline inherits from this last tradition directly. The open-source scientific computing ecosystem made the 20,221-test validation suite possible. Reference implementation: https://github.com/calebpruett927/GENERATIVE-COLLAPSE-DYNAMICS.

Appendix A: Heterogeneous Counterexample

To demonstrate why the integrity bound is a diagnostic and not merely an inequality, we contrast the homogeneous trace with two heterogeneous cases.

1. Moderate heterogeneity

Replace $\mathbf{c} = (0.99, 0.99, 0.99)$ with $\mathbf{c} = (0.85, 0.92, 0.98)$—the $t = 0$ state from the end-to-end reference casepack `UMCP-REF-E2E-0001`:

Output	Homogeneous	Heterogeneous
F	0.9900	0.9167
ω	0.0100	0.0833
κ	−0.01005	−0.08870
IC	0.9900	0.9151
Δ	0.0000	0.0015
S	0.0560	0.2665
C	0.0000	0.1062
Regime	Stable	Watch

The heterogeneous trace drops to Watch because $\omega = 0.083 > 0.038$ (drift gate fails). The gap appears: $\Delta = 0.0015$.

2. Geometric slaughter

The mechanism of *geometric slaughter* is demonstrated when a single channel approaches zero. Set $c_1 = 0.001$, $c_2 = c_3 = 0.99$:

$$\begin{aligned} F &= 0.6603 \quad \text{(mean still majority-healthy)}, \\ \mathrm{IC} &= 0.0993 \quad \text{(geometric mean destroyed)}, \\ \Delta &= 0.5610 \quad \text{(massive gap)}. \end{aligned}$$

One channel near ε kills IC because the geometric mean multiplies rather than averages. Fidelity F is the arithmetic mean: it is robust to a single outlier. Integrity IC is the geometric mean: it is sensitive to every channel simultaneously. This is why $\mathrm{IC} \leq F$ is not merely an inequality—it is a **diagnostic**. The gap Δ reveals what the arithmetic mean conceals.

Appendix B: Degenerate-Limit Derivations

1. Duality identity

The GCD duality $F+\omega = 1$ is definitional ($\omega := 1-F$). Classical unitarity $p + q = 1$ for a Bernoulli trial is the zero-cost special case where $\Gamma(\omega)$ is removed from the seam budget (20).

2. Integrity bound

For $n = 2$ equal-weight channels, $F = (c_1 + c_2)/2$ and $\mathrm{IC} = \sqrt{c_1 \cdot c_2}$. The condition $\mathrm{IC} \leq F$ is precisely $\sqrt{c_1 c_2} \leq (c_1 + c_2)/2$, i.e., the arithmetic–geometric mean inequality for two variables. For general n with weights, the integrity bound is strictly more general: it carries channel semantics (c_i represents a specific physical quantity), the guard band ($c_i \in [\varepsilon, 1 - \varepsilon]$), and composition laws (IC composes geometrically: $\mathrm{IC}_{12} = \sqrt{\mathrm{IC}_1 \cdot \mathrm{IC}_2}$; F composes arithmetically: $F_{12} = (F_1 + F_2)/2$).

The solvability interpretation: given (F, IC) with $n = 2$,

$$c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}, \tag{B1}$$

which requires $\mathrm{IC} \leq F$ for real solutions.

3. Bernoulli field entropy

The GCD entropy (11) is the full Bernoulli field entropy. Each channel c_i is a continuous variable in $(0, 1)$. When the collapse field is restricted to $c_i \in \{0, 1\}$ only, the binary entropy $h(c) = -c \log c - (1 - c) \log(1 - c)$ is evaluated only at the endpoints, and the sum reduces to Shannon's formula for a discrete source with outcome probabilities equal to the weights.

4. Log-integrity relation

$\mathrm{IC} = e^{\kappa}$ is definitional: $\kappa = \sum w_i \ln c_i$ defines κ, and $\mathrm{IC} := e^{\kappa}$. The classical exponential map $e^x : \mathbb{R} \to \mathbb{R}^+$ is what remains when the weighted sum is replaced by a scalar and the channel structure is removed.

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[2] C. Paulus, The physics of coherence: Recursive collapse & continuity laws (2025), pOST canon anchor; Weld-ID: W-2025-12-31-PHYS-COHERENCE.

[3] C. Paulus, Generative-collapse-dynamics (2026), reference implementation repository (GitHub).

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[6] Joint Committee for Guides in Metrology, Evaluation of measurement data — guide to the expression of uncertainty in measurement, JCGM 100:2008 (GUM 1995 with minor corrections) (2008), uncertainty expression standard; useful for UCD.MET pipeline semantics.